# Sparsh Dadhich

sparshdadhich@gmail.com | sdadhich04.github.io | linkedin.com/in/sparsh-dadhich | github.com/sdadhich04 | Seattle, WA

## Education

**University of Washington** Seattle, WA

*B.S. Electrical & Computer Engineering, Minor in Neural Computation & Engineering* *June 2026*

- Dean's List (multiple quarters); Annual Dean's List 2025–2026

## Experience

**Data Acquisition & Instrumentation Engineer** Jun. 2026 – Present

*ARC Lab, University of Washington (PI: Sep Makhsous)* *Seattle, WA*

- Lead the undergraduate team running the SHIELD stress-test campaign: assign test sessions, train students on the shaker-rig and Faraday-cage setups, and review all data before it enters the shared dataset
- Designed and built the Faraday cage and the automation scripts driving the orbital-shaker vibration-test rig: vibration, thermal, EMI and fault-injection testing for SHIBench, an open sensor-fault-detection benchmark
- Extended the team's dual-core ESP32-S3 FreeRTOS/ESP-IDF acquisition firmware (C/C++): noise-dataset logging across the IMU, microphone, pressure, temperature and photodiode drivers; two autonomous test sequencers
- Flagged a post-soak data-loss issue; produced sensor-characterization plots for QA; SHIBench manuscript co-author

**Biomedical Instrumentation & Impedance Lead** Sep. 2025 – Present

*UW BioRobotics Lab (PI: Blake Hannaford)* *Seattle, WA*

*Undergraduate Researcher* *Jun. 2024 – Sep. 2025*

- Built two bioimpedance instruments end to end for TPT-Finder, an intraoperative tissue-classification platform: the BioZ v4 tetrapolar board (AD5933, INA333, AD8606 Howland current source) and a two-electrode pen (AD5934)
- Calibrated BioZ v2 to 0.034% error at 33 kHz (2200 Ω); ran the study and all 500+ ex-vivo measurements
- Rewrote and documented the AD5933 I2C driver against the datasheet; diagnosed a sub-1 kΩ compression artifact
- Team won the inaugural SoundBio Lab Ignite Prize (67 teams), Hollomon Health Innovation Challenge, Mar. 2026

**Co-Hardware Lead & Procurement Lead, Project SHIELD** Jan. 2026 – Jun. 2026

*Guide Air Labs* *Seattle, WA*

- Co-led hardware (with Thomas Glass) for a data-acquisition device with 8 sensor types, including IMU, microphone, pressure, temperature and photodiode; ESP32-S3 DAQ PCB from schematic to fab
- Worked around a post-fab datasheet misread to avoid a respin; about $1,700 spent of a $4,000 budget (57% under)

**Computer Vision Robotics Intern** Aug. 2025 – Sep. 2025

*Central Research Laboratory, Bharat Electronics* *India*

## Projects

**BackOnTrack** | *ESP32-S3, C/C++, FreeRTOS, ESP-IDF, BNO085 IMU, EMG, BLE* 2026

- Wrote all the firmware (15/15 commits) for a four-person team's posture wearable: FreeRTOS/ESP-IDF rule-based classifier (good + 5 fault states), 95.3% accurate against real-time movements

**SmartCrib** | *Arduino Nano 33 BLE Sense, ESP32, TensorFlow Lite Micro, MFCC, C/C++* 2026

- Joint project with Jonathan Lu: PDM mic → MFCC → INT8 TFLM model fully on-device; ESP32 BLE alert bridge

**Real-time EMG BCI** | *MATLAB, Lab Streaming Layer, Kalman filter* 2026

- EMG streaming, Kalman-filter closed-loop cursor and feedback-lag sweep in a three-person EE 466 lab group (Winter 2026), following the course protocol and script outlines by A. L. Orsborn; my code and lag-sweep analysis

**EngineerGraph** | *Python, SQLite (FTS5, sqlite-vec), FastAPI* 2026

- Knowledge graph (7,170 nodes, 22,325 edges) linking 316 skills to evidence; cut a search from 1,604 SQL statements to 14 (0.4 s) with ranking proven identical; 702 tests

## Skills & Leadership

**Embedded**: C, C++, ESP32/ESP32-S3, FreeRTOS, ESP-IDF, Arduino, BLE, SPI, I2C, UART, KiCad, Onshape, Git
**Sensing & Signals**: AD5933/AD5934, INA333, Howland current sources, BNO085, MyoWare; MFCC, Kalman filters
**ML/TinyML**: TensorFlow Lite Micro, INT8 quantization, pruning; Python, PyTorch, scikit-learn, MATLAB
**Leadership**: Student Advisor, UW NextGen Civic Leader Corps (Dec. 2022 – Jun. 2026)